

Embedded systems exp 18

Program:

```
#include <reg51.h>

sbit PWM = P1^0;

unsigned char count = 0;
unsigned char duty = 100; // 0–200

void timer0_ISR(void) interrupt 1
{
    TH0 = 0xFC;          // 1 ms
    TL0 = 0x66;

    count++;

    if(count < duty)
        PWM = 1;
    else
        PWM = 0;

    if(count >= 200)
        count = 0;
}

void main()
{
```

```
TMOD = 0x01;          // Timer0, Mode 1
```

```
TH0 = 0xFC;
```

```
TL0 = 0x66;
```

```
IE = 0x82;            // EA=1, ET0=1
```

```
TR0 = 1;              // Start Timer0
```

```
while(1);
```

```
}
```

Output

